Mohammed Batis

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Page: https://batis1.github.io/batis1/
Date of Birth: 09/12/1998 Nationality: Yemeni

Research Interests

Reverse engineering artificial neural networks into human understandable algorithms.

My research interests lie in mechanistic interpretability and trustworthy AI. I aim to uncover the internal computational mechanisms of neural networks and large language models.

- Mechanistic interpretability of large language models
- Interpretable machine learning for biomedicine

Education

PhD Computer Science and Technology 2025 – Present
University of Science and Technology of China, Hefei, China

MSc Computer Science and Technology 05/10/2022 – 30/07/2025
Wenzhou University
Final grade: 4.5. Thesis: Research on the Improved RIME Optimization Algorithm and Its Application for Wrapper-based Feature Selection in Biomedical Data.

BSc Computer Science and Technology 07/10/2018 – 14/06/2022
Wenzhou University
Final grade: 4.03 GPA. Thesis: Design and Implementation of a Chinese HSK Test Preparation Platform.

Secondary School 14/08/2014 – 09/07/2017
Mukalla Model Secondary School
Final grade: 98.63%.

Publications

ACGRIME: Adaptive Chaotic Gaussian RIME Optimizer for Global Optimization and Feature Selection 2024
Cluster Computing Journal
Published. Article link.

Prey Capture Enhanced Harris Hawks Optimizer for Wrapper-based Feature Selection in High-Dimensional Medical Data 2024
Computer Methods and Programs in Biomedicine
Published. Article link.

Random differential RIME optimization for multi-threshold segmentation in dermoscopic skin cancer images 2024
Biomedical Signal Processing and Control
Under second revision.

An enhanced bat optimizer with elite selection and crisscross strategies for multi-threshold image segmentation of lupus nephritis 2024
International Journal of Computational Intelligence Systems
Under review.

Project

WHSK – Interactive HSK Preparation Platform 09/01/2022 – Current
Independent academic and software project
Developed a gamified web platform for HSK exam preparation, featuring interactive practice and progress tracking. Registered software copyright in China (No. 2025SR0444242). Project link: https://whsk.tech/.

Honours and Awards

- Chinese Government Scholarship at Wenzhou University (10/07/2022)
- Excellent Graduation Design Award at Wenzhou University (20/06/2022)
- Zhejiang Government Scholarship at Wenzhou University (09/10/2021)
- Second Place in Mukalla Model Secondary School Rankings (07/12/2017)
- 9th Best Middle School Graduate in the Province of Wadi Hadramout (04/07/2013)

Technical Skills

- Python, PyTorch, MATLAB
- React.js, Node.js
- MongoDB, MySQL
- Git and GitHub

Languages

- Arabic (mother tongue)
- English
- Chinese